ITT 036 | PROGRAMMING ASSIGNMENT 1

Digital Signal Encoder& Decoder

Project Specification Report

Student(s): 2024BITE019 2024BITE063 2024BITE068 • Semester: Autumn 2026

Project Overview

Dcom is a menu-driven C++ application for generating digital bitstreams, applying digital line codes, converting analog samples to Pulse Code Modulation (PCM), and visualizing signals. Its plotting interface uses GLFW to create a window and OpenGL to render waveforms and a reference grid.

Implemented Features

- Generate random bitstreams, force a run of four or eight consecutive zeros, or enter bits manually.
- Report the longest palindromic substring in the digital input.
- Encode using NRZ-L, NRZ-I, Manchester, Differential Manchester, AMI, B8ZS, or HDB3.
- Encode PCM samples as 3-bit indices across eight quantization levels; decode the bitstream using the supplied amplitude range.
- Plot encoded or reconstructed signals graphically.

Technology

The project uses C++ (C++17-compatible source) and the C++ Standard Library for strings, vectors, random generation, bit manipulation, and algorithms. GLFW and OpenGL provide signal visualization.

Assumptions & Limitations

Digital input is restricted to `0` and `1`. PCM applies eight uniform quantization intervals over the minimum-to-maximum input range and reconstructs each sample at its interval midpoint; each sample therefore uses three bits. Line-code waveforms use discrete signal levels. The supplied specification notes that PCM decoding is implemented, while the optional line-coding decoder (recovering bits from an encoded signal) is not included in the reviewed source.

Build & Run (macOS)

Install Xcode Command Line Tools and Homebrew, then install GLFW with `brew install glfw`. From the project directory, compile:

```
clang++ -std=c++17 main.cpp \
LineEncoding.cpp modulation.cpp \
demodulation.cpp plot.cpp \
-I/opt/homebrew/include \
-L/opt/homebrew/lib -lglfw \
-framework OpenGL -o dcom
```

Run with `./dcom`. Adjust include and library paths if Homebrew is installed outside `/opt/homebrew`.

References

Project source: `https://github.com/sHiNoBi005/Dcom`.
Assignment: ITT 036, Programming Assignment 1, 21 September 2026.
GLFW: `https://www.glfw.org/documentation.html`.
OpenGL: `https://www.opengl.org`.